EE4-50
EE9-FPN2-10

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2021

EEE PART IV: MEng and ACGI

SUSTAINABLE ELECTRICAL SYSTEMS

Thursday, 6 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer Question 1 & 4 in Booklet A, Questions 2 & 3 in Booklet B.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible	First Marker(s) :	B. Chaudhuri, G. Strbac
	Second Marker(s) :	G. Strbac, B. Chaudhuri

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Answer ALL Questions

1. Question on Wind and Photovoltaic Power

a) A doubly-fed induction generator with two back-to-back power converters on the rotor circuit is often used for variable speed wind turbines. If the rated power capacity of each power converter is 20% of the total power capacity of the generator, what range of variable speed operation (i.e. ratio of maximum to minimum rotor speed) is possible?

[3]

b) The hourly wind speed (in m/s) profile across the year at a particular location is given by the following Weibull probability distribution where the probability of experiencing wind speed less than v_w is given by:

$$\Phi(v_w) = 1 - e^{-\left(\frac{v_w}{8.5}\right)^{1.9}}$$

A wind farm at this location has 6 MW wind turbines with doubly-fed induction generators (DFIG). The wind turbine parameters are given in Table 1.1 and the density of air can be assumed to be 1.22 kg/m^3.

Length of turbine blades	70 m
Maximum power coefficient	0.46
Optimum tip speed ratio	7.8
Cut-in speed	3 m/s
Cut-out speed	26 m/s

Table 1.1: Parameters of the wind turbine

i) What is the probability of encountering wind speeds below the rated wind speed?

[2]

ii) What is the probability of encountering wind speeds at which the turbine doesn't extract any power?

[1]

iii) What is the probability of encountering wind speeds at which the wind turbine limits the power output to its rated power using pitch control?

[1]

iv) Determine the rated power capacity of each power converter of the DFIG (as a percentage of the total power capacity of the generator) that is needed to extract maximum power from wind over the relevant wind speed range.

[3]

c) The current-voltage characteristics of an ideal solar PV cell under 1000 W/ m^2 irradiance and 45°C is given by

$$I_{PV} = 4.1 - I_0\left[e^{\frac{qV_{PV}}{kT}} - 1\right]$$

where $q = 1.6 \times 10^{-19}$ C, $k = 1.39 \times 10^{-23}$ J/K, $I_0 = 1 \times 10^{-12}$ A

i) Calculate the open circuit voltage of the PV cell at 45°C.

[2]

ii) Calculate the short circuit current.

[1]

iii) Calculate the approximate power drawn by a 0.1 Ω resistor connected across the PV cell.

[1]

iv) Would the power drawn by a 1 Ω resistor across the PV cell be larger or smaller than that in part (iii).

[1]

d) A solar PV power park has 1000 PV modules each with a rated capacity of 350 W under standard test condition (STC). The surface area of each PV module is 1.35 m^2. The PV modules are laid out in the form of 50 parallel strings with each string having 20 PV modules connected in series with a string diode. The terminal voltage of each PV module is 65 V for 350 W output under STC.

i) What is the efficiency of each PV module under STC?

[2]

ii) What is the total power output (in kW) of the solar PV park when the irradiance is 750 W/m^2 assuming the same efficiency as under STC?

[1]

iii) How much power (in kW) is generated by the solar PV park under STC if 10 PV modules in separate strings are completely shaded?

[1]

iv) How much power (in kW) is generated by the solar PV park under STC if 10 PV modules within a single string are completely shaded?

[1]

2. Question on balancing, Voltage Control and Security of Supply

Part A

a) List alternative technologies that can provide reserve services for balancing of demand and supply.

[1.5]

b) Explain the advantages and disadvantages of using flexible technologies to offset the need for reserve provision from conventional generation.

[2.5]

Part B

c) What are the limiting factors and drivers for connecting wind and solar farms to distribution networks?

[1.5]

d) Describe different technologies and control techniques that can increase network hosting capacity for renewable generation?

[2.5]

Part C

Contribution of distributed generations to the security of supply depends on the demand profiles and availability of the resources. In this case, three units, connected to busbars A, B and C, respectively, can provide network support at busbar A as shown in Figure 2.1. Those units have ratings of 30, 20 and 10 MW and the availability of 90, 80 and 60%, respectively.

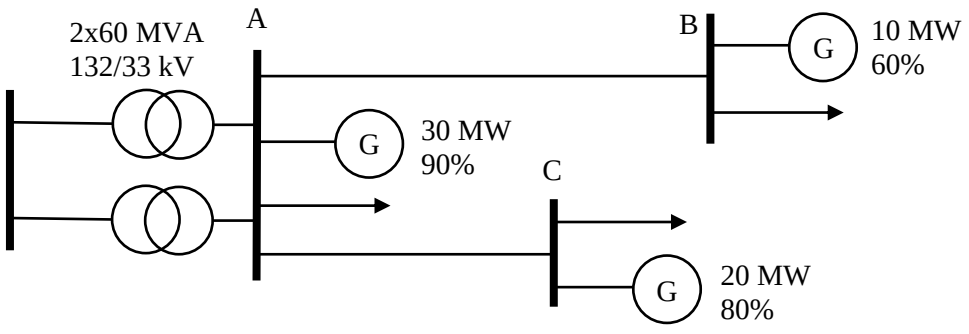


Figure 2.1 Diagram of the considered part of the network

Demand profile during the peak season that lasts for 100 days, is represented by the normalised load duration curve as shown in Figure 2.2.

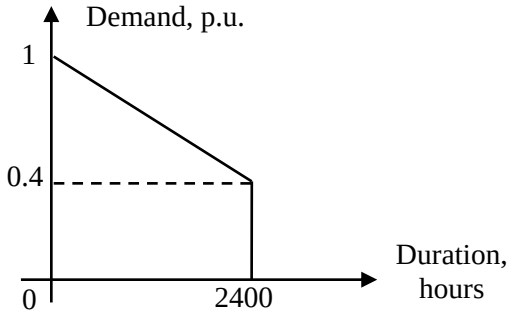


Figure 2.2 Normalised load duration curves for two characteristic days

e) Form the capacity outage probability table for plants providing network support at busbars A, and calculate the corresponding probability values.

[4]

f) Calculate the Expected Energy Not Supplied in the system considering the presence of generation units, using above load duration curve.

[5]

g) Calculate the firm capacity of the equivalent source that never fails, that results in the same Expected Energy Not Supplied.

[3]

3. Question on Low Carbon Heat

a) In 2008 the UK Government passed legislation entitled the Climate Change Act 2008. This required 80% reduction in the UK's greenhouse gas emissions (GHG) by 2050 compared to emissions in 1990. The Government also established the Committee on Climate Change (CCC).

i) Briefly summarise the purpose of the CCC.

[2]

ii) In December 2020 the CCC proposed the 6th Carbon Budget. Outline the key recommendations presented.

[2]

b) A building has a floor area of 100m^2 and an average thermal transmittance of 3 W_{th}/m^2K. Estimate the heat demand when the internal temperature is 20°C and the outside temperature is 7°C. State any assumptions made.

[3]

c) A householder is considering replacing their existing gas boiler with an air source heat pump (ASHP). Give a brief description of the operation of an ASHP.

[4]

d) The householder is trying to make a decision between buying a 5 kW_{th} or a 8.5kW_{th} ASHP. On average the winter temperature is 7°C but during a cold winter the temperature can drop to -5°C.

i) Using the same data listed in question B calculate the building space heat demand at -5°C.

[3]

ii) Estimate the internal building temperature if the householder selects the smaller heat pump.

[3]

iii) The 8.5 kW_{th} ASHP is much more expensive than the 5 kW_{th} ASHP and even though a cold winter does not occur very often and does not last long when it does, the householder does not want to be cold. The householder decides to improve the building's insulation. What percentage improvement in the building's average thermal transmittance is required to maintain an internal temperature of 20 °C during cold weather.

[3]

4. Question on Flexibility, Hydro Power and Offshore Transmission

Demand Flexibility

a) Explain key similarities and differences between using Demand-Side Response and energy storage to support power system operation and investment.

[2]

b) Heat demand diagram for a household during six contiguous peak demand hours on a cold winter day is presented in Figure 1. The household's heat demand is delivered by an air-source heat pump system with the hourly Coefficient of Performance (COP) values varying as shown in Figure 1. Answer the following questions:

i) What is the peak electricity demand for the household if we ignore any non-heating electricity consumption? Plot the household's electricity demand diagram during the six-hour period shown in Figure 4.1.

[2]

ii) By how much could you reduce the peak electricity demand using thermal energy storage (TES) with the heating power rating of 4 kW_{th} and energy rating of 7 kWh_{th}? Explain how TES should be used to achieve this peak demand reduction. Plot the household's electricity demand diagram after using TES. (Assume that the heat storage is fully charged at the beginning of Hour 1 and can be fully discharged by the end of Hour 6. Ignore any heat losses associated with storing and releasing heat from thermal storage.)

[6]

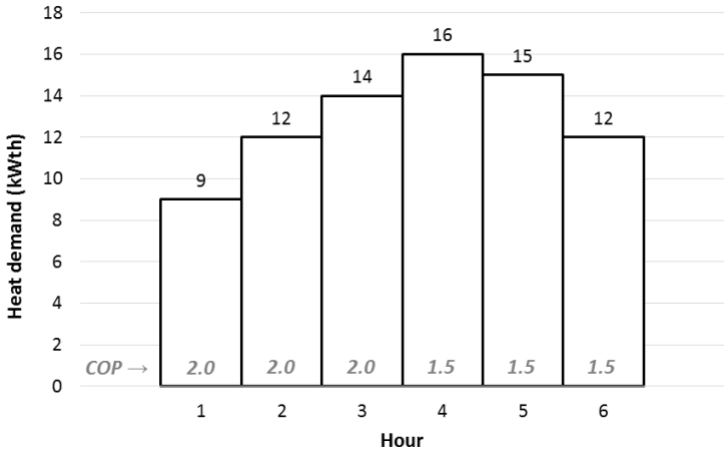


Figure 4.1. Household heat demand during peak demand period

Hydro Power

c) A reservoir based hydro power plant has a rated capacity of 1.0 GW with an overall efficiency of 92%. The maximum discharge of water is 1000 m^3/s. The density of water is 1000 kg/m^3, acceleration due to gravity is 9.81 m/s^2 and atmospheric pressure is 101.3 kPa.

i) If the height of the surface of the tailwater is 20 m, what should be the minimum height of the surface of the headwater in the reservoir to produce the rated power output?

[2]

ii) The inlet to the penstock is at a height of 50 m and the inner diameter of the penstock is 10 m. Calculate the velocity (m/s) and absolute pressure (kPa) of water at the inlet of the penstock for the condition in part (i) and 1000 m^3/s discharge.

[3]

Offshore Transmission

d) Justify with technical reasons the choice of the power transmission option (AC, LCC-HVDC or VSC-HVDC) for the following projects:

i) Western HVDC link in Great Britain with 250 miles of subsea cable.

[2]

ii) Connection to Hornsea One wind farm 120 km off the north-east coast of UK.

[2]

iii) Connection to BARD1 windfarm 100 km off the north coast of Germany.

[1]